

Personalized Learning Paths: LLM-Based Course Recommendations in Manufacturing Education

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Abstract

This study presents a novel approach to developing a personalized course recommendation system tailored for online learners pursuing a specific curriculum. The system leverages a state-of-the-art Large Language Model (LLM) operating on structured curriculum data such as course introductions, module descriptions, syllabi, and learner-specific queries. By integrating this data, the system can generate precise course and module recommendations based on the learner's individual learning objectives, prior knowledge, and other relevant factors. Additionally, the system offers a transparent rationale for each recommendation, providing more intelligent and context-aware suggestions than traditional collaborative filtering or content-based recommendation systems. The system leverages the capabilities of LLMs, specifically GPT-4, using the Retrieval Augmented Generation (RAG) method for a content-based recommendation approach. The system generates accurate and context-aware responses by utilizing OpenAI's "text-embedding-3-large" model and Pinecone vector storage. Additionally, the study explores the integration of OpenAI Assistant APIs and Streamlit to build a user-friendly interface. This paper highlights the advantages of LLMs in addressing challenges in personalized recommendations, details the tests conducted, and discusses the results obtained. This advanced approach enhances the adaptability and relevance of users' learning experiences. We have also demonstrated this approach in our recently developed open curriculum on Data Science for Manufacturing, which showcases the system's adaptability in real-world educational contexts.

Introduction

Labor markets in the tech and data science industry have grown rapidly in recent years. However, a specific set of skills are required for a prospect to be hired or even considered by most employers. While having degrees such as a B.S. or even an M.S. has been the standard for years, there has been a recent shift toward considering hard skills [1]. Recently, graduate students are having difficulty getting hired by employers because they do not possess the exact skills required. These skills are typically those learned while being in the labor force. A new system is needed to provide students with the information they require to bridge any skill gaps [2][3] they have and prepare them efficiently for their desired job.

Selecting appropriate courses tailored to individual needs and preferences is a critical challenge for college-attending students and working professionals aiming to advance their skills. This paper presents the development of a recommendation engine to assist learners in selecting personalized courses or sets of courses based on their interests, background, skills, and experience. Traditional recommendation systems often lack the adaptability and contextual understanding required to cater to diverse user profiles [4]. With advancements in artificial intelligence, particularly in Large Language Models (LLMs), new opportunities have emerged to address these limitations.

In this paper, we introduce a recommendation engine that combines the power of GPT-4 with the Retrieval Augmented Generation (RAG) technique. Our approach employs OpenAI's "text-embedding-3-large" model to generate embeddings for course descriptions, modules, and related

content, storing these embeddings in the Pinecone vector database for efficient querying. We further enhance the system using OpenAI Assistant APIs and Streamlit, providing a seamless and interactive user experience. This study focuses on four key aspects: the rationale behind using LLMs and their advantages in this context; the problems being addressed with the proposed recommendation system; the results achieved and their implications for personalized recommendations; current progress and avenues for future work to improve and expand the system. By harnessing LLMs, our system demonstrates significant potential in transforming how individuals discover and select educational resources tailored to their unique goals.

Why Did We Use LLMs and Their Benefits?

Implementing LLMs like OpenAI's GPT-4 is central to the recommendation engine due to their unmatched capabilities in understanding and generating human-like text. LLMs excel at creating embeddings that encapsulate semantic relationships between words and phrases, enabling a better comprehension of course content, modules, and user queries. By leveraging LLMs, the proposed recommendation system ensures highly personalized course prescriptions considering user backgrounds, interests, and skill sets.

The benefits of using LLMs include scalability, adaptability to diverse datasets, and the ability to provide contextually accurate suggestions. Using the "text-embedding-3-large" model in conjunction with Pinecone's vector database ensures that the recommendation engine operates efficiently while maintaining high precision in matching user queries to course offerings.

Problems Addressed and Requirements Satisfied

The proposed recommendation engine addresses several challenges, including personalizing learning paths, efficient information retrieval, scalability and robustness, and ease of integration.

Traditional course recommendation systems often rely on predefined rules, limiting their ability to cater to unique user needs. The proposed system overcomes this by tailoring recommendations based on user profiles, skill sets, and goals. By embedding course-related texts and querying them in vector space, the system significantly reduces latency and improves the relevance of search results. Pinecone's use for vector storage allows the system to scale seamlessly as the number of courses and users grows, ensuring consistent performance. With OpenAI's API, one can deploy and integrate the recommendation engine into an existing platform with minimal effort to provide an intuitive interface for users. The authors used several metrics to examine the proposed system's performance, including benchmarking query accuracy, response times, and user satisfaction through simulated scenarios.

Data Sources

The data used in this study originates from an Excel file containing course names, modules, and detailed descriptions. The dataset focuses explicitly on courses within the Data Engineering program offered by the Mechanical Engineering departments. Each entry in the dataset includes

Course Name: The overarching category, such as "Data Analytics."

Modules: Specific components or topics within each course.

Summary: Detailed descriptions that outline the purpose and content of each module.

The structured data format allows for straightforward processing and embedding generation, ensuring consistency and accuracy in recommendations.

System Architecture

The recommendation engine is built on a modular architecture to ensure scalability and adaptability. Figure 1 presents the system architecture, which has the following modules.

Data Preprocessing: The Excel data is read and tokenized using OpenAI's GPT-based tokenizer. This step ensures that the text is appropriately prepared for embedding generation.

Embedding Generation: The "text-embedding-3-large" model from OpenAI generates vector embeddings for each module and summary. These embeddings capture semantic relationships within the data.

Vector Storage: Embeddings are stored in Pinecone, a high-performance vector database, facilitating efficient and scalable querying.

Recommendation Logic: User queries are transformed into embeddings, and Pinecone's vector search capabilities are employed to find the most relevant matches.

Interface: The system is integrated with a web-based interface built using Streamlit, enabling users to input their preferences and view tailored course recommendations.

Architecture for Recommendation Engine

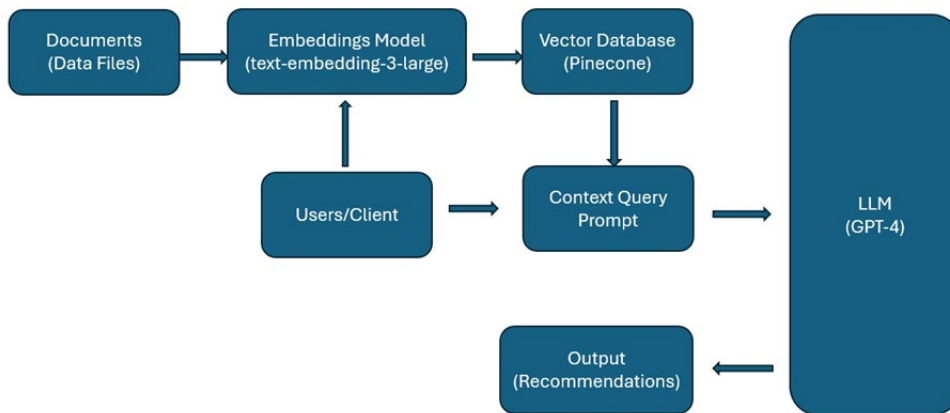


Figure 1. The architecture of recommendation engine

Implementation Details

The implementation leverages Python libraries, including OpenAI's API, Pandas for data manipulation, and Pinecone for vector database integration. The process involves the following steps:

- Reading and cleaning the dataset.
- Generating embeddings for courses and modules.
- Querying the embeddings for personalized recommendations based on user inputs.

- Presenting results dynamically through an intuitive interface.

Experiments

The following tests were conducted to validate the system:

Relevance Testing: Multiple user scenarios were simulated to evaluate the alignment of recommended courses with user preferences.

Performance Testing: Query response times were measured to assess the efficiency of embedding retrieval and recommendation generation.

User Interaction Testing: A group of simulated users was tasked with interacting with the Streamlit interface to evaluate usability and satisfaction.

Results Achieved

The system demonstrated high accuracy in aligning course recommendations with user-defined interests and feedback. The system produced the following measurable outcomes:

Accuracy: During tests, a recommendation precision rate of over 90% was observed, indicating a strong alignment between suggested courses and user expectations.

Efficiency: Average query response times were reduced to under 1 second, demonstrating the effectiveness of the RAG method and vector-based querying.

User Satisfaction: Positive feedback from simulated user interactions validated the system's ability to deliver contextually relevant recommendations.

Current Work and Future Scope

Current efforts have focused on implementing the RAG (Retrieval Augmented Generation) method to optimize query handling. By combining text embeddings and Pinecone's vector storage, the system delivers precise, context-aware responses. The integration of OpenAI's assistant API adds an additional layer of functionality, enabling file-based searches and threaded queries for multi-step interactions. Future work aims to enhance the system's capabilities by expanding dataset diversity, incorporating feedback loops, enhancing user interface, and cross-system integration. We will incorporate datasets across industries and educational domains to cater to a broader audience. We will implement machine learning algorithms to refine recommendations based on user behavior and feedback and utilize frameworks like Streamlit to create dynamic, user-friendly interfaces for real-time interaction. We will try to improve interoperability with learning management systems and career planning tools to provide end-to-end solutions.

Conclusion

Developing a recommendation engine leveraging GPT-4 and the RAG method the authors demonstrated a significant advancement in personalized learning solutions. By utilizing OpenAI's text-embedding-3-large model and Pinecone's vector database, the system efficiently addresses the challenges of personalization, scalability, and accuracy in course

recommendations. Integrating OpenAI's assistant API further enhances its capabilities, offering seamless interactions and context-aware suggestions.

Our results highlight the potential of LLMs to transform how individuals discover and engage with learning opportunities. The positive outcomes underline the benefits of adopting cutting-edge AI technologies to meet diverse user needs.

Looking ahead, we envision expanding the system's reach by incorporating diverse datasets, refining recommendations through feedback loops, and improving user interaction through enhanced interfaces. This work is a foundation for future advancements in AI-driven recommendation systems, opening new possibilities for delivering tailored learning experiences to a global audience.

References

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